



Real-time time-dependent DFT

Updates on the development in ABINIT

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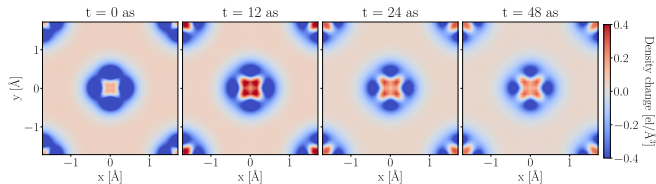
ABINIT developers workshop 2026

☐☐☐ Sant Feliu de Guíxols

10 June 2026

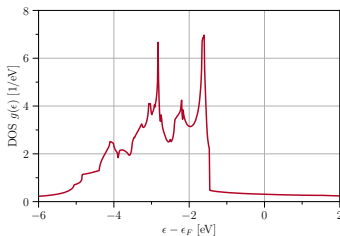


$$i\partial_t\psi_{nk}(\underline{r},t) = \hat{H}[n](\underline{r},t)\psi_{nk}(\underline{r},t)$$

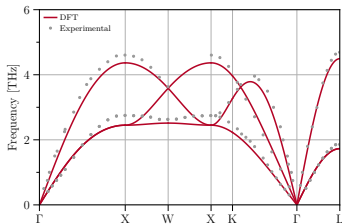


Time-dependent properties from DFT

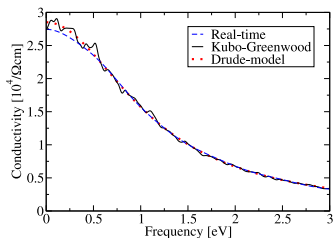
- **DFT** : ground-state properties (electron density, energy, ...)
- **Perturbation theory (DFPT)** : *linear* and *non-linear* response to *static* perturbations
Phonons, Raman spectra, dielectric tensor ...
- **Linear response theory** : *linear* response to *static* or *dynamic* perturbations (From ground-state KS orbitals)
Kubo-Greenwood formula for transport coefficients or optical properties
- **Time-dependent DFT (TDDFT)** : time-dependent properties (excited states)
Excitation energies, optical properties, electronic spectroscopy ...



DOS of FCC Copper at $T_e = 500$ K.



Phonons frequencies of FCC Gold at $T_e = 500$ K compared to inelastic neutron scattering from [1]



Electronic conductivity of liquid Al at $\rho = 2.7$ g/cc & $T \approx 5800$ K [2]

Time-dependent DFT (TDDFT)

Runge-Gross & Van Leeuwen – *Hohenberg-Kohn & Kohn-Sham for TD systems*

$$v(\underline{r}, t) \xleftrightarrow{\Psi_0} n(\underline{r}, t)$$

The expectation value of any observable is a functional of the *time-dependent* electron density $n(\underline{r}, t)$ and the *initial state* Ψ_0

$$\langle \Psi[n, \Psi_0](t) | \hat{O} | \Psi[n, \Psi_0](t) \rangle \equiv O[n, \Psi_0]$$

$n(\underline{r}, t)$ can be obtained as stationary point of the action

$$A[n, \Psi_0] = \int_{t_0}^{t_1} \langle \Psi[n, \Psi_0] | i\partial_t - \hat{H}(t) | \Psi[n, \Psi_0] \rangle dt$$

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■ Introducing an effective system of independent electrons $\Rightarrow$ Time-dependent Kohn-Sham equations

$$i\partial_t \psi_{nk}(\underline{r}, t) = \hat{H}[n](\underline{r}, t) \psi_{nk}(\underline{r}, t)$$

$$\hat{H}[n](\underline{r}, t) = -\frac{\nabla^2}{2} + \int \frac{n(\underline{r}', t)}{|\underline{r} - \underline{r}'|} d\underline{r}' + v_{xc}[n](\underline{r}, t) + v(\underline{r}, t) \quad \text{and} \quad n(\underline{r}, t) = \sum_{n,k} f_{nk}^0 |\psi_{nk}(\underline{r}, t)|^2$$

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■ Exchange-correlation potential is a functional of the electron density *at all previous times*!

$$v_{xc}[n](\underline{r}, t) \equiv v_{xc}[n(\{t'\}), t' \leq t](\underline{r}, t)$$

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- Exchange-correlation potential is a functional of the electron density *at all previous times!*

$$v_{xc}[n](\underline{r}, t) \equiv v_{xc}[n(t)](\underline{r}, t)$$

Adiabatic approximation
Local in time – ignore memory effect!

Real-time TDDFT (RT-TDDFT)

Time-dependent Kohn-Sham equations

$$\begin{aligned}i\partial_t\psi_{nk}(\underline{r}, t) &= \hat{H}(t)\psi_{nk}(\underline{r}, t) \\ \psi_{nk}(\underline{r}, t) &= \hat{U}(t, t_0)\psi_{nk}(\underline{r}, t_0)\end{aligned}$$

Direct numerical integration :

$$\begin{aligned}\psi_{nk}(\underline{r}, t + \Delta t) &= \hat{U}(t + \Delta t, t)\psi_{nk}(\underline{r}, t) \\ \psi_{nk}(\underline{r}, t + \Delta t) &= \exp\left[-i\int_t^{t+\Delta t}\hat{H}(t')dt'\right]\psi_{nk}(\underline{r}, t) \\ \psi_{nk}(\underline{r}, t + \Delta t) &\approx \exp\left[-i\hat{H}(t + \Delta t/2)\Delta t\right]\psi_{nk}(\underline{r}, t)\end{aligned}$$

To get the time evolution of KS orbitals and the electron density :

$$n(\underline{r}, t) = \sum_{n,k} f_{nk}^0 |\psi_{nk}(\underline{r}, t)|^2$$

RT-TDDFT in the projector augmented wave (PAW) approach

TDKS equations in PAW

$$i\hat{S}\partial_t |\tilde{\psi}_{nk}(t)\rangle = [\hat{\tilde{H}}(t) + \hat{\tilde{P}}] |\tilde{\psi}_{nk}(t)\rangle$$

$$\hat{\tilde{H}}(t) = \hat{T}^\dagger \hat{H}(t) \hat{T}$$

$$\hat{\tilde{P}} = -i\hat{T}^\dagger \partial_t \hat{T} = 0 \quad \text{for fixed nuclei}$$

PAW

$$|\psi\rangle = \hat{T} |\tilde{\psi}\rangle$$

$$\hat{T} = 1 + \sum_{ai} \left[|\phi_{ai}\rangle - |\tilde{\phi}_{ai}\rangle \right] \langle \tilde{p}_{ai}|$$

$$\hat{S} = \langle \tilde{\psi} | \tilde{\psi} \rangle = 1 + \sum_{aij} |\tilde{p}_{ai}\rangle \left[\langle \phi_{ai} | \phi_{aj} \rangle - \langle \tilde{\phi}_{ai} | \tilde{\phi}_{aj} \rangle \right] \langle \tilde{p}_{aj}|$$

Direct numerical integration (fixed nuclear positions) :

$$|\psi_{nk}(t + \Delta t)\rangle = U(t + \Delta t, t) |\psi_{nk}(t)\rangle$$

$$|\psi_{nk}(t + \Delta t)\rangle = \exp \left[-i\hat{S}^{-1} \int_t^{t+\Delta t} H(t') dt' \right] |\psi_{nk}(t)\rangle$$

$$|\psi_{nk}(t + \Delta t)\rangle \approx \exp \left[-i\hat{S}^{-1} H(\underline{r}, t + \Delta t/2) \Delta t \right] |\psi_{nk}(t)\rangle$$

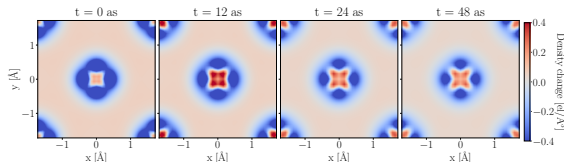
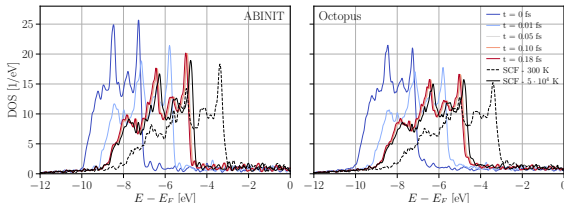
Implementation in ABINIT

$$\tilde{\psi}_{nk}(\underline{r}, t + \Delta t) = \exp \left[-i S^{-1} \tilde{H}(\underline{r}, t + \Delta t/2) \Delta t \right] \tilde{\psi}_{nk}(\underline{r}, t)$$

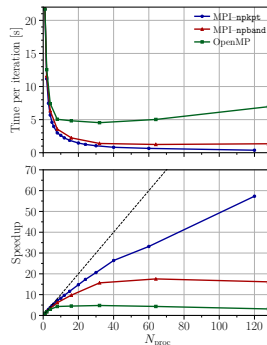
Propagator : $e^{-i S^{-1} \tilde{H} \Delta t} = \sum_n \frac{(-i S^{-1} \tilde{H})^n}{n!} + \text{Predictor-corrector scheme}$



Tests & Tutorials available



Parallel MPI / OpenMP



Implementation in ABINIT

Location in source files : `src/95_drive/m_rttdfft_driver.F90` \& `src/80_rttdfft/*.F90`

Typical input

```
optdriver 9      # RT-TDDFT
getwfk_filepath "/path/to/initial/wfk" # Initial orbitals
dtele 0.01      # time step
ntime 5000      # number of propagation step
td_exp_order 4  # Order of the Taylor expansion of the exponential
td_propagator 1 # Propagator (0: ER, 1: EMR)
td_scnmax 4     # Max number of corrector steps
td_scthr 1e-8   # Convergence threshold for corrector steps
td_restart 0    # restart (1: restart 0: from scratch)
etc.
```

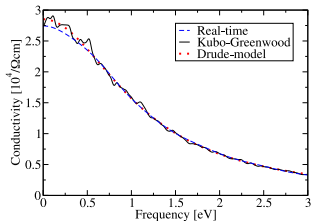
Typical output

Ni.abo	- Main output file	Nio_TDCURRENT	- Current density file (prtcurrent 1)
TD_RESTART	- Restart file	Nio_TIME_OCC	- Occupations at TIME (prtocc 1)
Nio_TDENER	- Energy file	Nio_TIME_DOS	- DOS at TIME (prtdos 1)
etc.			

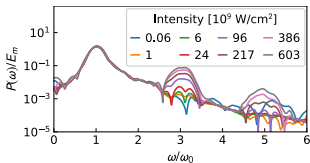
Some applications of RT-TDDFT

Response functions

Within the linear regime & beyond



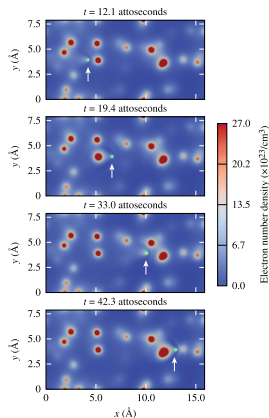
Conductivity of liquid Al [1]



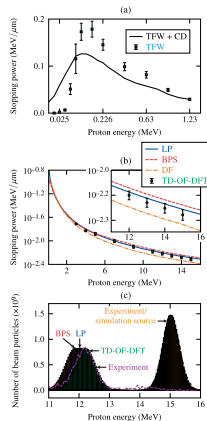
High-harmonic generation in solid Si [2]

Out-of-equilibrium processes

For instance in matter under irradiation

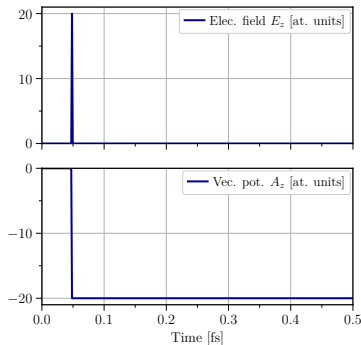


Stopping power of protons in a Be plasma [3]



- [2] Andrade, Hamel & Correa *EPJB* 91 229 (2018)
 [2] Rodrigues Pela & Draxl *Electron. Struct.* 3 037001 (2021)
 [3] Ding *et al.* *PRL* 121 145001 (2018)

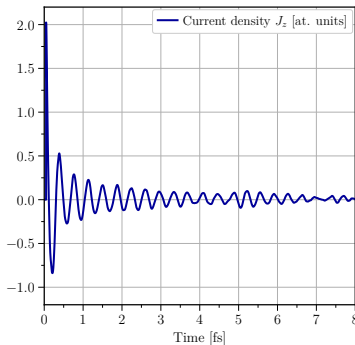
Optical properties from RT-TDDFT



Impulse electric field

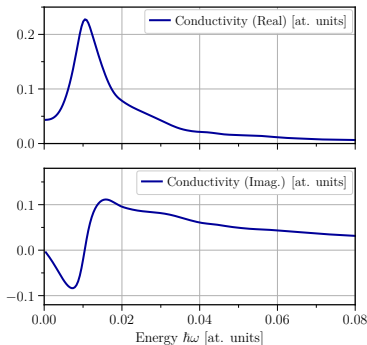
$$\underline{E}(t) = E_0 \delta(t - t_0) \underline{e}_x$$

$$\underline{A}(t) = E_0 \theta(t - t_0) \underline{e}_x$$



Macroscopic induced current density

$$J_z(t) = -\frac{1}{\Omega} \text{Im} \left[\sum_{nk} f_{nk}^0 \langle \psi_{nk}(t) | \nabla_z | \psi_{nk}(t) \rangle \right] - \frac{N A_z(t)}{\Omega}$$



Optical conductivity

$$\tilde{J}_z(\omega) = \tilde{\sigma}_{zz}(\omega) \tilde{E}_z(\omega)$$

Dielectric function

$$\tilde{\epsilon}_{zz}(\omega) = 1 + \frac{4\pi i \tilde{\sigma}_{zz}(\omega)}{\omega}$$

RT-TDDFT with external electric field

Long wavelength approximation

$$\underline{E}(\underline{r}, t), \underline{A}(\underline{r}, t) \Rightarrow \underline{B} = \underline{\nabla} \times \underline{A} = \underline{0}$$

Scalar ϕ and vector $\underline{A}$ potentials

$$\underline{E} = -\underline{\nabla}\phi - \partial_t \underline{A}, \underline{B} = \underline{\nabla} \times \underline{A}$$

Gauge invariance The $\underline{E}$ and $\underline{B}$ fields remain invariant through a gauge transformation

$$\phi' = \phi - \partial_t \chi(\underline{r}, t), \underline{A}' = \underline{A} + \underline{\nabla} \chi(\underline{r}, t)$$

$$\psi'(\underline{r}, t) = \exp(-i\chi(\underline{r}, t)) \psi(\underline{r}, t) = \hat{T}_\chi(\underline{r}, t) \psi(\underline{r}, t)$$

Length gauge

$$\phi(\underline{r}, t) = \underline{E}(t) \cdot \underline{r}, \underline{A} = \underline{0}$$

$$H[n](\underline{r}, t) = -\frac{\nabla^2}{2} + v[n](\underline{r}) - \underline{E}(t) \cdot \underline{r}$$

Velocity gauge

$$\phi = 0, \underline{A}(t) = -\int_0^t \underline{E}(t') dt'$$

$$H[n](\underline{r}, t) = \frac{1}{2}(-i\underline{\nabla} + \underline{A}(t))^2 + v[n](\underline{r})$$

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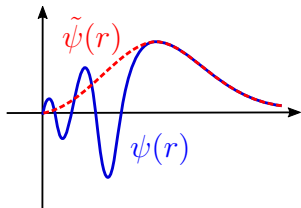
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Gauge invariance and PAW (Time-dependent Gauge-including PAW – TD-GIPAW)

Pseudo-potential – PAW



$$|\psi\rangle = \hat{T} |\tilde{\psi}\rangle$$

Gauge invariant PAW transformation $|\psi\rangle = \hat{T}_{\underline{A}} |\tilde{\psi}\rangle$

$$\hat{T}_{\underline{A}}(\underline{r}, \underline{r}', t) = e^{-i\underline{A}(t) \cdot \underline{r}} \hat{T}(\underline{r}, \underline{r}') e^{i\underline{A}(t) \cdot \underline{r}'}$$

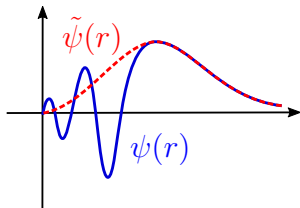
$$\hat{T}_{\underline{A}}(\underline{r}, \underline{r}', t) = 1 + \sum_{\alpha} e^{-i\underline{A}(t) \cdot \underline{r}} \left(\phi_{\alpha}(\underline{r}) - \tilde{\phi}_{\alpha}(\underline{r}) \right) \tilde{p}_{\alpha}^{*}(\underline{r}') e^{i\underline{A}(t) \cdot \underline{r}'}$$

$$\hat{T}_{\underline{A}}(\underline{r}, \underline{r}', t) = 1 + \sum_{\alpha} \left(\phi_{\alpha}^{\underline{A}}(\underline{r}) - \tilde{\phi}_{\alpha}^{\underline{A}}(\underline{r}) \right) \tilde{p}_{\alpha}^{\underline{A}}(\underline{r}')^{*}$$

$\Rightarrow \tilde{p}_{aij}^{\underline{A}}, \rho_{ij}^{\underline{A}}$ and $D_{ij}^{\underline{A}}$ now depend on the vector potential $\underline{A}(t)$.

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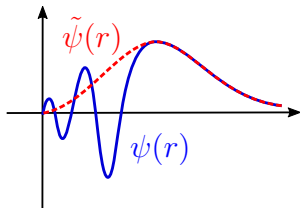
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With $\hat{\underline{P}}_{\underline{A}} = -i\hat{T}_{\underline{A}}^{\dagger}\partial_t\hat{T}_{\underline{A}}$

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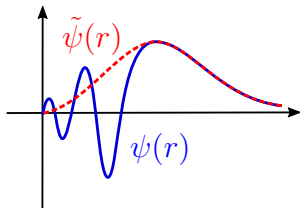
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Impulse electric field $\Rightarrow$ constant vector potential $\Rightarrow \hat{P}_{\underline{A}} = 0$

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$\Rightarrow \tilde{p}_{aij}^A, \rho_{ij}^A$ and D_{ij}^A now depend on the vector potential $\underline{A}(t)$.

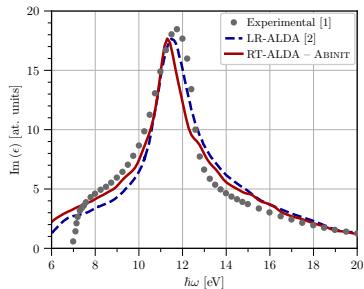
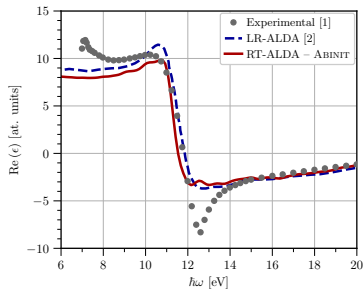
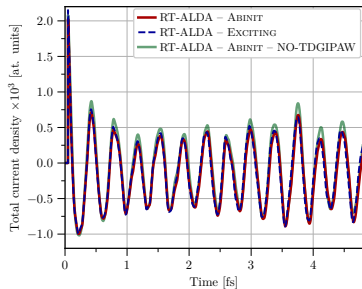
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Impulse electric field $\Rightarrow$ constant vector potential $\Rightarrow \hat{P}_{\underline{A}} = 0$

Macroscopic current density

$$\underline{J}(t) = -\frac{1}{\Omega} \text{Im} \left[\sum_{nk} f_{nk}^0 \langle \tilde{\psi}_{nk}(t) | \nabla | \tilde{\psi}_{nk}(t) \rangle + \sum_{\alpha} \rho_{\alpha}^A \left(\langle \phi_{\alpha}^A(t) | \nabla | \phi_{\alpha}^A(t) \rangle - \langle \tilde{\phi}_{\alpha}^A(t) | \nabla | \tilde{\phi}_{\alpha}^A(t) \rangle \right) \right] - \frac{\underline{A}(t)}{\Omega} \int_{\Omega} \tilde{n}(\underline{r}, t) d\underline{r}$$

Dielectric function of diamond



[1] Edwards & Philipp *Handbok of optical constants of solids* (1985)

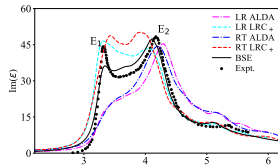
[2] Botti *et al. PRB* 69 155112 (2004)

The Exciting code : Gulans *et al. J. Phys. Cond. Matt.* 26 363202 (2014)

Current status & limitations

- RT-TDDFT not yet compatible with all possible options in ABINIT
spin polarization, SOC, DFT+U ...
- External electric field only compatible with PAW
- “Real” time-dependent electric field not yet implemented (beyond Dirac pulse)

- Adiabatic approximation. Can we go beyond?
Long-range kernel, TDCDFT [1]



Dielectric function of Si from RT-TDDFT with a “long-range kernel” from [1]

- Simulations of large boxes is difficult due to the large number of states to evolve
GPUs and/or better OpenMP parallelism could help
- Going to high-temperatures is also hard due to high energy states
Combining with the extended approach (extFPMD) ?

Conclusions and perspectives

- Time evolution of KS orbitals is available
- Response to Dirac pulse electric field is also implemented
 - Some questions remain (DC conductivity, vector potential induced by current, beyond linear-response)
- Generalizing the implementation to electric field pulse with different shape
 - Requires implementing the additional $\hat{P}_{\underline{A}}$ term
- Computation of other response functions ?
- Coupling with nuclear motion (MD) to go beyond the BO approximation (Ehrenfest dynamics)

Acknowledgments : L. Baguet & M. Torrent